

AUTONOMOUS KINETIC DEFENSE PLATFORM

R.A.S.K. 01 // KEEPER SERIES

Robotic Assistance, Security, Keeper — Field Operator Manual

*“Efficiency in the field is a matter of design.”*

System Overview

The **R.A.S.K. 01** (Robotic Assistance, Security, Keeper) is a 2.20-meter heavy-duty autonomous kinetic defense platform. Engineered for emergency extraction, high-risk civil pacification, and active search-and-rescue, it combines dynamic quadrupedal kinematics with an integrated life-support rescue capsule.

Key Functional Highlights

- **Kinematic Locomotion:** Heavy quadrupedal hydraulic articulation with gyroscopic self-righting algorithms.
- **Situational Awareness:** 360-degree spherical dome with dual LiDAR, thermal FLIR, and acoustic mapping.
- **High-Speed Pursuit:** Interception sprint speeds up to 25 km/h.
- **Extended Mission Profile:** High-density 45 kWh energy bank delivering 48 hours of continuous operation.
- **Armor Matrix:** Certified Level IV composite plates with deployable high-absorption riot shield.
- **Rear Extraction Unit:** Sealed life-support rescue cockpit for human casualty evacuation.

Dimensional & Operational Specifications

| Parameter                | Specification                   | Operational Context                                  |
|--------------------------|---------------------------------|------------------------------------------------------|
| Total Height             | 2.20 m                          | Imposing tactical presence and cockpit clearance     |
| Shoulder Width           | 1.60 m                          | Structural stability for ballistic shield deployment |
| Chassis Depth            | 1.50 m                          | Enclosed sealed life-support capsule compartment     |
| Dry Mass                 | 480 kg                          | Anti-rollover ballast and blast shock absorption     |
| Chassis Material         | Al-Alloy + Hardened Steel       | Multi-layer ballistic and chemical resistance        |
| Degrees of Freedom       | 16 (8 per arm, 4 per leg)       | Multi-directional terrain and obstacle navigation    |
| Power Plant              | Li-Ion Solid-State 48V / 45 kWh | Hermetically isolated core battery bank              |
| Operational Autonomy     | 48 Hours (2 Days)               | Uninterrupted tactical perimeter monitoring          |
| Maximum Sprint Velocity  | 25 km/h                         | High-speed pursuit and tactical interception         |
| Standard Patrol Velocity | 8 km/h                          | Formation pacing alongside human personnel           |
| Payload Capacity         | 220 kg                          | Adult casualty extraction or heavy cargo conveyance  |

## Schematic Overview & Blueprints

Below are the primary orthogonal projections extracted from the master assembly schematic.

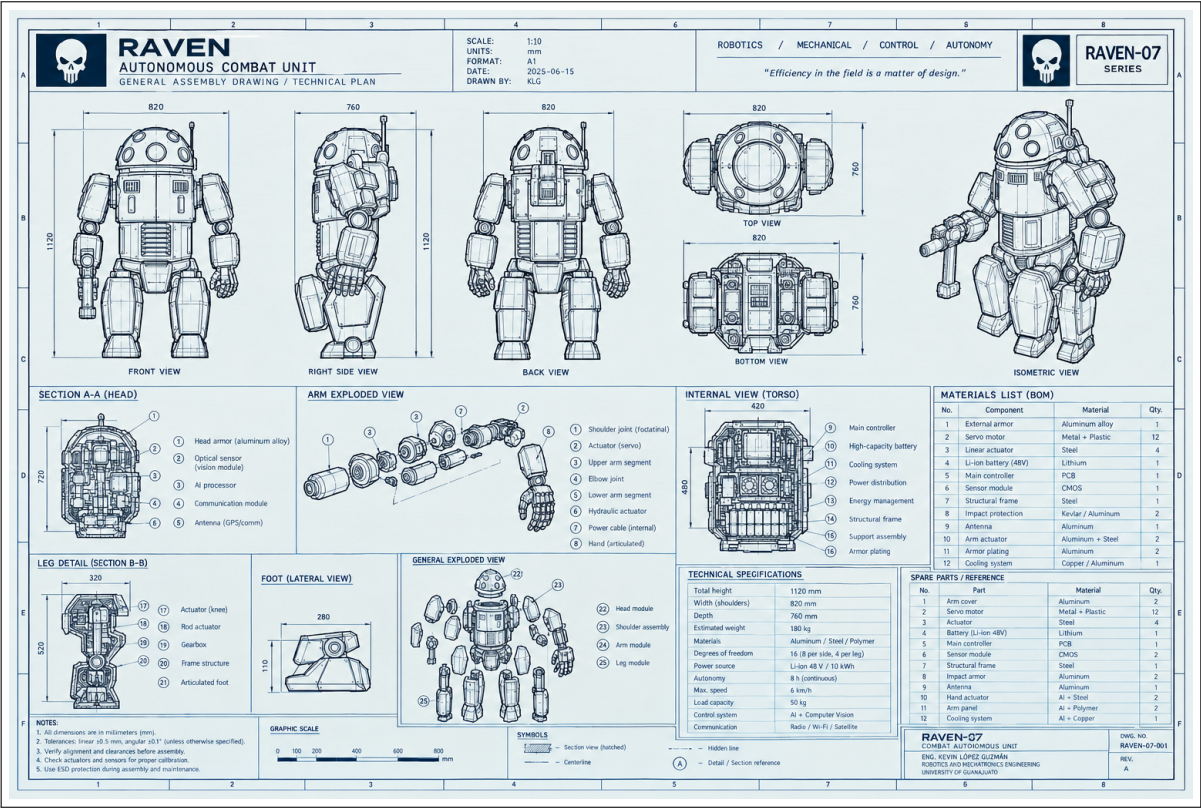


Figure 1: Master Blueprint and Dimensional Schematics (Scaled to 2.20 m Architecture).

## Bill of Materials Summary

| Item No. | Sub-Assembly             | Material / Standard                | Quantity        |
|----------|--------------------------|------------------------------------|-----------------|
| 01       | External Armor Shell     | 7075-T6 Hardened Alloy Plates      | 1 Outer Hull    |
| 02       | High-Torque Servomotors  | Industrial Steel Geared Assemblies | 12 Units        |
| 03       | Linear Leg Actuators     | Heavy Reinforced Carbon Steel      | 4 Units         |
| 04       | Energy Storage Bank      | 45 kWh Solid-State Li-Ion Cells    | 1 Internal Pack |
| 05       | Main ECU & AI Controller | Triple-redundant embedded PCB      | 1 Unit          |
| 06       | Optical & LiDAR Array    | CMOS Cluster + Solid-State LiDAR   | 2 Sensor Pods   |
| 07       | Internal Frame           | High-Tensile Structural Steel      | 1 Core          |
| 08       | Articulated Hand Limbs   | High-Strength Alloy + Nitrile Grip | 2 Assemblies    |
| 09       | Sealed Cooling System    | Dual Closed-Loop Fluid Cooling     | 1 Unit          |

## Safety Protocols for Field Personnel

- Safe Approach Distance:** Maintain at least 2.0 meters clearance during dynamic sprinting or stair ascension.
- Emergency Neutralization:** A manual disconnect lever is located behind the upper dorsal maintenance hatch.

3. **Lethal Force Protocol:** Escalation beyond non-lethal deterrents strictly requires remote two-factor human authentication.